

Aggregation

1. What is aggregation in Python?
2. What is the difference between aggregation and inheritance?
3. What is a HAS-A relationship?
4. How can aggregation be implemented in Python?
5. What is the difference between aggregation and composition?

Method Overriding

1. What is method overriding?
2. Why is method overriding used?
3. What happens when a child class defines a method with the same name as the parent?
4. How can you call the parent method from an overridden child method?
5. What is the role of super() in method overriding?

Method Overloading

1. What is method overloading?
2. Does Python support traditional method overloading?
3. Why doesn't Python support traditional method overloading?
4. How can you achieve method-overloading-like behavior in Python?
5. How can default arguments be used for method overloading?

Encapsulation

1. What is encapsulation?
2. Why do we use encapsulation?
3. How is encapsulation implemented in Python?
4. What are public, protected, and private members?
5. What is the difference between _variable and __variable?
6. What is name mangling in Python?
7. What are getter and setter methods?
8. Why are getter and setter methods used with encapsulation?

Instance & Class/Static Variables

1. What is an instance variable?
2. What is a static variable?
3. What is the difference between an instance variable and a class variable?
4. Where is an instance variable stored and where is a class variable defined?
5. Can an instance variable have a different value for every object?

Instance & Static Methods

1. What is an instance method?
2. What is a static method?
3. Why is self used in an instance method?
4. Why doesn't a static method require self?
5. What is the difference between an instance method and a static method?
6. When should you use a static method?
7. Can a static method access instance variables directly?